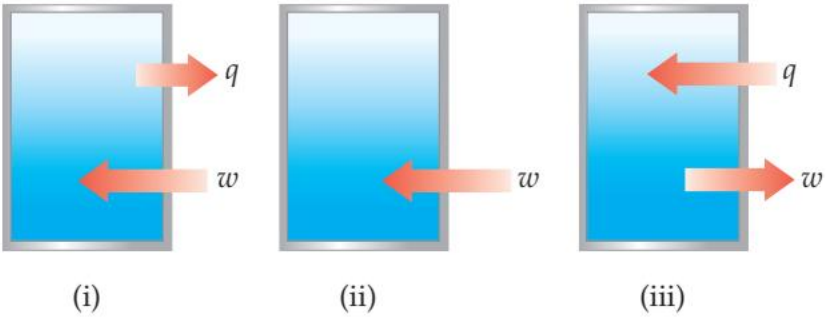
Chapter 05 Homework

Part A multiple choice (select the one that is best in each case. 1 point/question)

E. 1. Objects can possess energy as _____.
(a) endothermic energy
(b) potential energy
(c) kinetic energy

A) a only B) b only C) c only D) a and c E) b and c

A. 2. The contents of the closed box in each of the following illustrations represent a system, and the arrows show the changes to the system during some process. The lengths of the arrows represent the relative magnitudes of q and w. Which of these processes is both exothermic and $\Delta E > 0$?



$\Delta E = q + w.$

$w = -P\Delta V.$

A) (i) B) (ii) C) (iii) D) (i) and (ii) E) (ii) and (iii)

D. 3. The internal energy of a system _____.
A) is the sum of the kinetic energy of all of its components
B) is the sum of the rotational, vibrational, and translational energies of all of its components
C) refers only to the energies of the nuclei of the atoms of the component molecules
D) is the sum of the potential and kinetic energies of the components
E) none of the above

E. 4. Which one of the following is true concerning enthalpy? $H = E + PV.$
A) It is measured under conditions of constant volume.
B) The enthalpy change of a reaction is the reciprocal of the enthalpy change of the reverse reaction.
C) It usually is largely different in magnitude than the internal energy change of a process.
D) It is an intensive property, always having the units energy per mole.
E) It is a state function.

D 5. A chemical reaction that gives off heat to its surroundings is said to be _____ and has a _____ value of ΔH.
A) endothermic, positive
B) exothermic, positive
C) endothermic, negative
D) exothermic, negative

D. 6. From the following enthalpies of reaction,
$H_2(g) + F_2(g) \rightarrow 2\ HF(g)$ $\Delta H = -537$ kJ
$C(s) + 2\ F_2(g) \rightarrow CF_4(g)$ $\Delta H = -680.$ kJ
$2\ C(s) + 2\ H_2(g) \rightarrow C_2H_4(g)$ $\Delta H = +52.3$ kJ
calculate ΔH for the reaction of ethylene with F_2:
$C_2H_4\ (g) + 6\ F_2(g) \rightarrow 2\ CF_4(g) + 4\ HF(g)$
A) −2382 kJ B) −1165 kJ C) −1269 kJ D) −2486 kJ E) +234 kJ

E. 7. $A(g) + B(g) \rightarrow C(g)$. As the reaction occurs, the system loses 1.15 kJ of heat to the surroundings. As the volume of the gas decreases under the constant pressure of the atmosphere, the surroundings do 480 J of work on the system. What is the change in the internal energy of the system?

$\Delta E = q + w$

$H = E + pV.$

A) 1630 J B) 670 J C) −1630 J D) 478 J E) −670 J

D. 8. Which of the following quantities—heat, work, enthalpy, internal energy, pressure, volume—depend only on the end points and not on the path followed in a process?
A) enthalpy and internal energy
B) internal energy
C) heat
D) enthalpy, internal energy, pressure, volume
E) work

SF: $E = q + w$, $H = E + pV$, p, V.

PF: q, w.

A. 9. A volume of 50.0 mL of 0.400 *M* HBr at 24.35 ℃ is added to 50.0 mL of 0.400 *M* NaOH, also at 24.35 ℃. The final temperature is 27.06 ℃) Calculate the enthalpy change, Δ*H*, in kJ for the following reaction:
HBr(*aq*) + NaOH(*aq*) → NaBr(*aq*) + H_2O(*l*)
(~~Heat capacity of the~~ system is 0.418 kJ/°C)

$\Delta H = -q$

A) −1.13 kJ $q = C \cdot \Delta T$. B) −56.6 kJ C) −28.4 kJ D) 56.6 kJ E) 1.13 kJ

A. 10. The combustion of methylhydrazine (CH_6N_2), a liquid rocket fuel, produces N_2(*g*), CO_2(*g*), and H_2O(*l*):
2 CH_6N_2(*l*) + 5 O_2(*g*) → 2 N_2(*g*) + 2 CO_2(*g*) + 6 H_2O(*l*)
When 5.52 g of methylhydrazine is combusted in a bomb calorimeter, the temperature of the calorimeter increases from 25.00 to 45.00 ℃. In a separate experiment the heat capacity of the calorimeter is measured to be 7.794 kJ/℃. Calculate the heat of reaction for the combustion of a mole of CH_6N_2.
A) -1.30×10^3 kJ/mol
B) −156 kJ/mol
C) 1.30×10^3 kJ/mol
D) 156 kJ/mol
E) −18.72 kJ/mol

C. 11. If the heat of formation of H_2O(*l*) is −286 kJ/mol, which of the following thermochemical equations is correct?
A) 2 H(*g*) + O(*g*) → H_2O(*l*) Δ*H* = −286 kJ
B) 2 H_2(*g*) + O_2(*g*) → 2 H_2O(*l*) Δ*H* = −286 kJ
C) H_2(*g*) + 1/2 O_2(*g*) → H_2O(*l*) Δ*H* = −286 kJ
D) H_2(*g*) + O(*g*) → H_2O(*g*) Δ*H* = −286 kJ
E) H_2O(*l*) → H_2(*g*) + 1/2 O_2(*g*) Δ*H* = −286 kJ

$= q$.
$= q + (-p\Delta V) + p\Delta V$
$= q + w + p\Delta V$

$\Delta E = q + w$. $\Delta H = \Delta E + p\Delta V$.

A. 12. A gas is confined to a cylinder under constant atmospheric pressure. When the gas undergoes a particular chemical reaction, it absorbs 824 J of heat from its surroundings and has 0.650 kJ of P–V work done on it by its surroundings. What are the values of Δ*E* and Δ*H* for this process?
A) 1474J, 824J B) 824J, 0.650kJ C) 174J, 824J D) 174J, 174J E)1474J, 0.650kJ

824.

C. 13. Given 2 SO_2(*g*) + O_2(*g*) → 2 SO_3(*g*), which of the following equations is correct at 1atm and 298K?
A) $\Delta H_f^o(SO_3) = \Delta H^o_{rxn} - \Delta H_f^o(SO_2)$
B) $\Delta H_f^o(SO_3) = \Delta H^o_{rxn} + \Delta H_f^o(SO_2)$
C) $2\Delta H_f^o(SO_3) = \Delta H^o_{rxn} + 2\Delta H_f^o(SO_2)$
D) $2\Delta H_f^o(SO_3) = \Delta H^o_{rxn} - 2\Delta H_f^o(SO_2)$
E) $2\Delta H_f^o(SO_3) = 2\Delta H_f^o(SO_2) - \Delta H^o_{rxn}$

C. 14. Which of the following processes are endothermic?
A) burning a math B) freezing ice cream
C) melting chocolate D) cooling hot water
E) condensing water vapor

B. 15. Which substance undergoes the greatest temperature change when the same mass of each substance absorbs the same quantity of heat?

A) $CH_4(g)$ (Cs=2.20 J/g·K) B) $CaCO_3(s)$ (Cs=0.82 J/g·K)
C) $H_2O(l)$ (Cs=4.18 J/g·K) D) $CO_2(g)$ (Cs=0.84 J/g·K)

Part B: Short Answer -- Write legibly and show all work for all steps in the problem. (15 points)

16. The decomposition of sodium bicarbonate (baking soda), $NaHCO_3(s)$, into $Na_2CO_3(s)$, $H_2O(l)$, and $CO_2(g)$ at constant pressure requires the addition of 85 kJ of heat per two moles of $NaHCO_3$. (**a**) Write a balanced thermochemical equation for the reaction. (**b**) Draw an enthalpy diagram for the reaction.

17. Diethyl ether, $C_4H_{10}O(l)$, a flammable compound that has long been used as a surgical anesthetic, has the structure $H_3C\text{-}CH_2\text{-}O\text{-}CH_2\text{-}CH_3$ The complete combustion of 1 mol of $C_4H_{10}O(l)$ to $CO_2(g)$ and $H_2O(l)$ yields $\Delta H^o = -2723.7$ kJ. **(a)** Write a balanced equation for the combustion of 1 mol of $C_4H_{10}O(l)$. **(b)** By using the information in this problem and data below, calculate ΔH_f^o for diethyl ether.

Substance	ΔH_f^o (kJ/mol)
$CO_2(g)$	−393.5
$O_2(g)$	0
$H_2O(g)$	−241.82
$H_2O(l)$	−285.83

18. Consider the following reaction:

$2\ Mg(s) + O_2(g) \rightarrow 2\ MgO(s)$ $\Delta H = -1204$ kJ

(**a**) Is this reaction exothermic or endothermic?
(**b**) Calculate the amount of heat transferred when 3.55 g of Mg(*s*) reacts at constant pressure.
(**c**) How many grams of MgO are produced during an enthalpy change of −234 kJ?
(**d**) How many kilojoules of heat are absorbed when 40.3 g of MgO(*s*) is decomposed into Mg(*s*) and$O_2(g)$ at constant pressure?

19. Suppose that the gas-phase reaction $2\ NO(g) + O_2(g) \rightarrow 2\ NO_2(g)$ were carried out in a constant-volume container at constant temperature. (**a**) Would the measured heat change represent ΔH or ΔE? (**b**) If there is a difference, which quantity is larger for this reaction? (**c**) Explain your answer to part (b).

Chapter 05 Homework Answer Sheet

Name: 陈.陈 **Student ID:** 12311410 **Instructor:**_____ **Score:**_____

Part A multiple choice (15 points).

1-5 EADED

6-10 DEDAA

11-15 CACCB

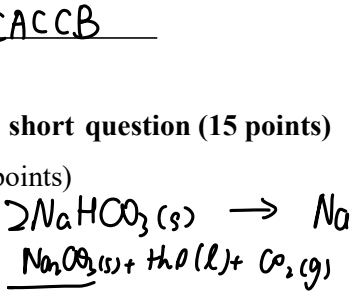
16. The decomposition of sodium bicarbonate (baking soda), $NaHCO_3(s)$, into $Na_2CO_3(s)$, $H_2O(l)$, and $CO_2(g)$ at constant pressure requires the addition of 85 kJ of heat per two moles of $NaHCO_3$. (**a**) Write a balanced thermochemical equation for the reaction. (**b**) Draw an enthalpy diagram for the reaction.

Part B short question (15 points)

16. (4 points)

(a) $2NaHCO_3(s) \rightarrow Na_2CO_3(s) + H_2O(l) + CO_2(g)$ $\Delta H = +85$ kJ

(b)



17. (4 points)

(a) $C_4H_{10}O(l) + 6O_2(g) \rightarrow 5H_2O(l) + 4CO_2(g)$ $\Delta H = -2723.7$ kJ

(b) $\Delta H_{rxn} = 5\Delta H_f(H_2O(l)) + 4\Delta H_f(CO_2(g)) - \Delta H_f(C_4H_{10}O(l)) - 6\Delta H_f(O_2(g))$

$\Rightarrow \Delta H_f(C_4H_{10}O(l)) = -279.5$ kJ/mol.

18. (4 points)

(a). exothermic.

(b) $n(Mg) = \frac{3.55g}{24.305 g/mol} = 0.146 mol$ $q = 0.146 mol \times \frac{-1204 kJ}{2 mol} = -87.9$ kJ

(c) $\frac{234}{1204} mol \times 2 \times 40.305 g/mol = 15.7 g.$

(d) $\frac{40.3g}{40.3g/mol} \div 2 \times [-(-1204 kJ/mol)] = 602$ mol

19. (3 points)

(a) ΔE.

(b) $\Delta E = q$.

$\Delta H = \Delta E + \Delta(pV) = q + \Delta(nRT) = q + \Delta n \cdot RT = q - RT.$

$\Rightarrow \Delta H < \Delta E.$

(c) explained as above already.